

Tyler Meng: Statement of Teaching Interests

I believe that an effective teacher in the geosciences should be adaptable in leveraging the diversity of learning styles, background knowledge, interests, and ideas brought to campus by their students with the objective of focusing them on a central set of principles, questions, problems, or goals. Earth and environmental sciences include a wide range of subdisciplines and impact a variety of stakeholders; therefore, the students who are drawn to study Geology and Geophysics will also come with a diverse set of specific experiences, values, and aspirations for their societal contributions stemming from an education at the University of Wyoming. As a teacher, one of the most important skills that I will implement is continuing education and learning for myself; each term brings a unique set of students whose backgrounds and goals may differ significantly, particularly for graduate coursework. Not only would I be diligent in preparing coursework with up-to-date material each term, but I would also actively assess the growth of the students through multiple forms of measurement. I would employ a combination of traditional lectures, problem sheet assignments, written exams, audiovisual presentations, and research-based term projects that allow students to collect data, think critically, and directly practice combining the skills necessary to produce professional deliverables.

My previous teaching experience includes two semesters as a Teaching Assistant in the Lunar and Planetary Laboratory at the University of Arizona. For both semesters, I assisted in teaching the “Universe and Humanity” course, which serves as an introduction to space sciences for non-geoscience students. For this course, I facilitated in-class activities, provided detailed feedback to students with their graded work, and held weekly office hours to provide additional assistance to the students. This experience gave me a deeper understanding of the range of background knowledge, areas of interest, and preferred methods for synthesizing information over a diverse class of students. It was extremely rewarding to watch the growth of the students who curiously asked questions and returned to office hours, and these conversations allowed me to practice different methods of communicating the concepts to a broad audience. In addition to my teaching assistantships, I have also delivered guest lectures to undergraduate and graduate-level courses focused on the Solar System and Radar Remote Sensing. My ability to connect with a diverse audience and adapt my communications via real-time feedback will directly benefit my teaching of undergraduate and graduate courses, and it will allow me to train more early-career scientists and science communicators to effectively work to distribute scientific information within their respective communities.

In addition to my experience as a teaching assistant and guest lecturer, I have contributed to the mentorship of several undergraduates and graduate students researching environmental science, and I have facilitated field-based learning experiences for students to directly collect impactful geophysical data as a part of their geological field studies curricula. In summer 2022, as a PhD candidate, I served as a guest instructor for the University of Arizona’s “Geophysical Field Methods for Planetary Analogs” course, where we brought enrolled first year graduate students to a rock glacier field site in Wyoming to learn about the geophysical methods applicable to alpine permafrost. Not only did this course provide a first-time geophysical field methods experience for the students, but it also resulted in an increased volume of data that was subsequently published with the participating students as co-authors. This demonstrates my ability to inclusively involve students in active research while also teaching them the fundamentals of the cryosphere and geophysical field methods. I am continuing to mentor a student from this cohort in their research into drone-based geophysics in the alpine cryosphere.

This past summer, in August 2025, I organized a collaborative effort involving undergraduates, graduate students, and professors from Washington University in St. Louis, Colorado School of Mines, University of Alaska Fairbanks, and University of Colorado Colorado Springs. For this campaign, which totaled 23 individuals, we visited a rock glacier in the San Juan Mountains of southern Colorado and collected ground-penetrating radar data, terrestrial laser scanning data, and drone-based photogrammetry data. This expedition served as a foundational field geophysics experience for the undergraduates in the group and provided the opportunity for graduate students to acquire data with various instruments while considering their thesis topics and research questions. This real-world experience will continue to benefit these students as we work to process, visualize, and eventually communicate our results using the data that they contributed to. One of my graduate student mentees is working to interpret the GPR data that we collected, while another is working to develop data processing algorithms to remotely sense the active layer thickness over the rock glacier. I deeply value the transfer of knowledge that stems from a successful mentorship, and I believe that a teacher learns and grows alongside their students. For example, I am serving as a mentor to an undergraduate researcher who is studying the role of wildfire in permafrost thaw dynamics, and I hope to expand upon this knowledge to mentor future students who may express similar interests.

My background is strongly suited to teach several courses offered in the Geology undergraduate program, including Geological Remote Sensing, Geohydrology, Earth Surface Processes, and Glaciology. I would also be interested in developing a “Planetary Geophysics” course for both graduates and undergraduates that builds upon the foundations of Remote Sensing and covers the wide array of processes that occur in our solar system while exploring the physical laws and essential parameters governing such processes. This course would consist of written assessments, group discussions, and a data analysis project that encourages students to generate research questions and consider how these surface processes may affect society and our understanding of the natural universe. This course would accompany the existing educational resources in the Department of Geology and Geophysics while offering a unique perspective on the diverse surface processes driving change throughout the solar system, and it would allow students to apply their prerequisite knowledge and critical thinking skills to include a wide range of geological settings.

Ultimately, I believe my role as an educator is to contribute to society through different forms of communication. The obvious form of communication is the direct transfer of my knowledge to the next generation of geoscientists. More subtly, I aim to share my deep appreciation for the Earth and the sciences that allow our communities to interact meaningfully with the world around us. By sharing this passion and working with students to foster their growth into independent scientists and conscientious citizens of Earth, I hope that they will propagate this appreciation for the environment throughout society. My goal as an educator is to provide the most inclusive space possible to promote a diversity of ideas while fostering curiosity and scientific literacy in students who hope to positively impact society.